

DSC540WeekProjectMilestone5

June 1, 2025

1 Project Milestone 5

```
[546]: # Importing Pandas packages required for this exercise.
```

```
import sqlite3
import pandas as pd
import seaborn as sns
import matplotlib

import urllib.request, urllib.parse, urllib.error
import requests
from bs4 import BeautifulSoup
import ssl
import re

import matplotlib.dates as mdates
from matplotlib.dates import DateFormatter
import matplotlib.ticker as tick
```

```
[548]: conn = sqlite3.connect("DSC540FinalProjectDB", timeout=10)

cursor = conn.cursor()
```

```
[550]: try:
    with sqlite3.connect("DSC540FinalProjectDB.db") as conn:
        cursor = conn.cursor()
        conn.commit()
        print("Succeeded")
except sqlite3.OperationalError as e:
    print("Failed to open database:", e)
```

Succeeded

```
[552]: # loading covid CSV file for data transformation and cleaning

covid = pd.read_csv("./time_series_covid_19_confirmed.csv")
covid = covid.rename(columns={'Country/Region': 'Country'})
```

```
covid.head()
```

```
[552]:
```

	Province/State	Country	Lat	Long	1/22/20	1/23/20	1/24/20	\
0	NaN	Afghanistan	33.0000	65.0000	0	0	0	
1	NaN	Albania	41.1533	20.1683	0	0	0	
2	NaN	Algeria	28.0339	1.6596	0	0	0	
3	NaN	Andorra	42.5063	1.5218	0	0	0	
4	NaN	Angola	-11.2027	17.8739	0	0	0	

	1/25/20	1/26/20	1/27/20	...	4/4/20	4/5/20	4/6/20	4/7/20	4/8/20	\
0	0	0	0	...	299	349	367	423	444	
1	0	0	0	...	333	361	377	383	400	
2	0	0	0	...	1251	1320	1423	1468	1572	
3	0	0	0	...	466	501	525	545	564	
4	0	0	0	...	10	14	16	17	19	

	4/9/20	4/10/20	4/11/20	4/12/20	4/13/20
0	484	521	555	607	665
1	409	416	433	446	467
2	1666	1761	1825	1914	1983
3	583	601	601	638	646
4	19	19	19	19	19

[5 rows x 87 columns]

```
[554]: # Selecting 4 countries that has highest number of cases

covidMelt = covid.melt(id_vars = ['Country'])
covidDropNan = covidMelt.dropna()
covidDropVariable = covidDropNan[covidDropNan['variable'] != 'Province/State']
covidCountries = covidDropVariable.groupby(['Country'])['value'].sum().
    .reset_index().sort_values(by = 'value',ascending = False)
covidCountries = covidCountries.rename(columns={'value': 'Covidcases'})
covidCountries[0:5]
```

```
[554]:
```

	Country	Covidcases
171	US	6278184.3773
36	China	5267388.7564
84	Italy	2977275.0
156	Spain	2574218.0
65	Germany	1986377.0

```
[556]: # Write the covid countries DataFrame to a SQLite table
covidCountries.to_sql("covidCountries", conn, if_exists="replace")
```

```
[556]: 185
```

```
[558]: # reading covidtopCountries table data in sqlite database
res = conn.execute("SELECT * FROM covidCountries")
for row in res:
    print(row)
```

```
(171, 'US', 6278184.3773)
(36, 'China', 5267388.7564)
(84, 'Italy', 2977275.0)
(156, 'Spain', 2574218.0)
(65, 'Germany', 1986377.0)
(61, 'France', 1748435.0816000002)
(80, 'Iran', 1329967.0)
(175, 'United Kingdom', 923997.1052)
(170, 'Turkey', 526623.20699999999)
(161, 'Switzerland', 445000.0457)
(90, 'Korea, South', 390965.0)
(16, 'Belgium', 379608.8333)
(120, 'Netherlands', 362341.0139)
(32, 'Canada', 283292.1318)
(9, 'Austria', 248548.0663)
(23, 'Brazil', 228935.8397)
(135, 'Portugal', 212170.1754)
(83, 'Israel', 150816.0)
(160, 'Sweden', 147645.0)
(126, 'Norway', 128255.9409)
(138, 'Russia', 125270.0)
(8, 'Australia', 112560.8928)
(82, 'Ireland', 105561.4503)
(46, 'Denmark', 99170.84909999999)
(45, 'Czechia', 90898.2905)
(86, 'Japan', 86311.0)
(35, 'Chile', 85974.7819)
(78, 'India', 82647.0)
(105, 'Malaysia', 79299.0)
(51, 'Ecuador', 79100.9854)
(134, 'Poland', 78867.06450000001)
(137, 'Romania', 73497.91)
(128, 'Pakistan', 65871.72039999999)
(132, 'Peru', 61137.7948)
(133, 'Philippines', 59545.0)
(101, 'Luxembourg', 55244.9449)
(145, 'Saudi Arabia', 51259.0)
(79, 'Indonesia', 49532.132)
(165, 'Thailand', 43873.0)
(60, 'Finland', 43185.0)
(47, 'Diamond Princess', 41300.0)
(111, 'Mexico', 41241.081699999995)
(174, 'United Arab Emirates', 38103.0)
```

(147, 'Serbia', 37518.0224)
 (67, 'Greece', 37226.8984999999996)
 (136, 'Qatar', 36082.5387000000005)
 (129, 'Panama', 35916.7559000000004)
 (150, 'Singapore', 35136.1166)
 (154, 'South Africa', 33442.378)
 (50, 'Dominican Republic', 33230.5730000000004)
 (77, 'Iceland', 31339.9423)
 (37, 'Colombia', 30892.2736)
 (6, 'Argentina', 27772.9672)
 (52, 'Egypt', 26103.0)
 (173, 'Ukraine', 25373.545)
 (2, 'Algeria', 24044.6935)
 (42, 'Croatia', 23376.3)
 (152, 'Slovenia', 22376.1467)
 (56, 'Estonia', 22022.6089)
 (81, 'Iraq', 20519.0)
 (116, 'Morocco', 19245.6990999999998)
 (12, 'Bahrain', 19019.5775)
 (121, 'New Zealand', 18123.9853999999998)
 (15, 'Belarus', 17185.6632)
 (112, 'Moldova', 16394.7815)
 (76, 'Hungary', 15526.6657999999999)
 (7, 'Armenia', 15190.1073)
 (100, 'Lithuania', 14942.0507)
 (92, 'Kuwait', 14367.25)
 (21, 'Bosnia and Herzegovina', 12729.5950000000001)
 (96, 'Lebanon', 12526.717)
 (10, 'Azerbaijan', 11723.7200000000001)
 (151, 'Slovakia', 11192.368)
 (25, 'Bulgaria', 11035.2197)
 (88, 'Kazakhstan', 11030.9432999999999)
 (95, 'Latvia', 10601.4828)
 (125, 'North Macedonia', 10344.3539)
 (169, 'Tunisia', 10298.0)
 (3, 'Andorra', 9758.0281)
 (163, 'Taiwan*', 9479.7)
 (31, 'Cameroon', 9391.3501)
 (40, 'Costa Rica', 9321.9955)
 (44, 'Cyprus', 8656.5563)
 (176, 'Uruguay', 8135.7114)
 (177, 'Uzbekistan', 7746.9628)
 (143, 'San Marino', 7340.4002)
 (1, 'Albania', 7158.3216)
 (87, 'Jordan', 7008.75)
 (127, 'Oman', 7002.0)
 (26, 'Burkina Faso', 6972.6767)
 (0, 'Afghanistan', 6828.0)

(43, 'Cuba', 6581.0)
(179, 'Vietnam', 6265.0)
(41, 'Cote d'Ivoire', 5837.9929)
(108, 'Malta', 5435.3129)
(66, 'Ghana', 5285.9233)
(75, 'Honduras', 4716.9581)
(146, 'Senegal', 4548.045)
(180, 'West Bank and Gaza', 4539.1854)
(124, 'Nigeria', 4221.7573)
(110, 'Mauritius', 4201.3)
(13, 'Bangladesh', 4154.0413)
(123, 'Niger', 4034.6895)
(157, 'Sri Lanka', 3948.0)
(64, 'Georgia', 3905.6723)
(93, 'Kyrgyzstan', 3777.9705)
(115, 'Montenegro', 3657.8)
(24, 'Brunei', 3559.263)
(178, 'Venezuela', 3375.8341)
(20, 'Bolivia', 3292.1211000000003)
(91, 'Kosovo', 3049.5056130000003)
(39, 'Congo (Kinshasa)', 2952.7204)
(30, 'Cambodia', 2884.4667)
(89, 'Kenya', 2466.8826)
(70, 'Guinea', 2152.249)
(168, 'Trinidad and Tobago', 2070.4692999999997)
(139, 'Rwanda', 1988.9336)
(131, 'Paraguay', 1916.1136999999999)
(99, 'Liechtenstein', 1749.69)
(48, 'Djibouti', 1709.4153999999999)
(113, 'Monaco', 1492.15)
(69, 'Guatemala', 1419.5527)
(103, 'Madagascar', 1390.1022)
(53, 'El Salvador', 1157.8977)
(85, 'Jamaica', 1088.8121)
(167, 'Togo', 1058.4443)
(14, 'Barbados', 977.6507)
(107, 'Mali', 949.574526)
(58, 'Ethiopia', 912.6347000000001)
(172, 'Uganda', 905.0)
(183, 'Zambia', 722.8666000000001)
(38, 'Congo (Brazzaville)', 679.7203999999999)
(106, 'Maldives', 614.4235)
(164, 'Tanzania', 536.5198)
(27, 'Burma', 526.8722)
(62, 'Gabon', 526.8057)
(72, 'Guyana', 505.25)
(55, 'Eritrea', 500.9617)
(114, 'Mongolia', 497.7092)

(11, 'Bahamas', 495.63800000000003)
 (18, 'Benin', 403.62350000000004)
 (73, 'Haiti', 376.686)
 (59, 'Fiji', 373.35159999999996)
 (162, 'Syria', 372.79889000000003)
 (54, 'Equatorial Guinea', 365.5)
 (97, 'Liberia', 363.9986)
 (71, 'Guinea-Bissau', 361.6233)
 (94, 'Laos', 352.351766)
 (98, 'Libya', 327.56343100000004)
 (119, 'Nepal', 319.4167)
 (148, 'Seychelles', 299.8124)
 (118, 'Namibia', 292.5328)
 (158, 'Sudan', 282.0804)
 (153, 'Somalia', 277.3517)
 (117, 'Mozambique', 258.863867)
 (4, 'Angola', 249.6712)
 (5, 'Antigua and Barbuda', 233.2644)
 (19, 'Bhutan', 227.9478)
 (57, 'Eswatini', 221.9434)
 (49, 'Dominica', 221.04399999999998)
 (34, 'Chad', 213.1864)
 (74, 'Holy See', 208.3563)
 (184, 'Zimbabwe', 208.0)
 (141, 'Saint Lucia', 201.9305)
 (130, 'Papua New Guinea', 170.6405)
 (159, 'Suriname', 165.8915)
 (33, 'Central African Republic', 165.5505)
 (68, 'Grenada', 161.4375)
 (109, 'Mauritania', 160.9487)
 (166, 'Timor-Leste', 145.853322)
 (29, 'Cabo Verde', 135.497)
 (102, 'MS Zaandam', 125.0)
 (140, 'Saint Kitts and Nevis', 115.574824)
 (22, 'Botswana', 114.35640000000001)
 (104, 'Malawi', 114.047217)
 (63, 'Gambia', 100.1331)
 (17, 'Belize', 76.6507)
 (149, 'Sierra Leone', 75.680666)
 (142, 'Saint Vincent and the Grenadines', 72.6971)
 (28, 'Burundi', 72.5458)
 (182, 'Yemen', 68.069115)
 (155, 'South Sudan', 63.184)
 (181, 'Western Sahara', 51.3297)
 (122, 'Nicaragua', 44.658199999999994)
 (144, 'Sao Tome and Principe', 38.799441)

```
[560]: # loading covid CSV file for data transformation and cleaning
url = 'https://en.wikipedia.org/wiki/COVID-19_pandemic_by_country_and_territory'
```

```
[562]: dfs = pd.read_html(url)
```

```
[563]: corona_df=dfs[12]
corona_df_col=corona_df.drop(columns=['Unnamed: 0'])
```

```
[564]: # remove records where deaths are equals to 0 or less
corona_df_clean=corona_df_col[(corona_df_col['Deaths'])>"0"]
```

```
[565]: # Write the covid DataFrame to a SQLite table
corona_df_clean.to_sql("coronawebdata", conn, if_exists="replace")
```

```
[565]: 232
```

```
[570]: def read_json_from_web_api(url):
    """
    Reads a JSON file from a web API.

    Args:
        url (str): The URL of the web API endpoint.

    Returns:
        dict: The JSON data as a Python dictionary, or None if an error occurs.
    """
    try:
        response = requests.get(url)
        response.raise_for_status() # Raise HTTPError for bad responses (4xx
        ↪ or 5xx)
        return response.json()
    except requests.exceptions.RequestException as e:
        print(f"Request error: {e}")
        return None
    except json.JSONDecodeError as e:
        print(f"JSON decode error: {e}")
        return None
```

```
[572]: # call Covid 19 API and load it in Json Format
url = "https://api.covidtracking.com/v1/us/daily.json" # Replace with the
        ↪ actual API endpoint
data = read_json_from_web_api(url)
df = pd.json_normalize(data)
df = df.drop('hash',axis=1)
df.head(10)
```

[572]:

	date	states	positive	negative	pending	hospitalizedCurrently	\
0	20210307	56	28756489.0	74582825.0	11808.0	40199.0	
1	20210306	56	28714654.0	74450990.0	11783.0	41401.0	
2	20210305	56	28654639.0	74307155.0	12213.0	42541.0	
3	20210304	56	28585852.0	74035238.0	12405.0	44172.0	
4	20210303	56	28520365.0	73857281.0	11778.0	45462.0	
5	20210302	56	28453529.0	73590280.0	11196.0	46388.0	
6	20210301	56	28399281.0	73334501.0	11748.0	46738.0	
7	20210228	56	28351189.0	73216424.0	11708.0	47352.0	
8	20210227	56	28296840.0	73012825.0	11731.0	48871.0	
9	20210226	56	28225595.0	72807735.0	11945.0	51112.0	

	hospitalizedCumulative	inIcuCurrently	inIcuCumulative	\
0	776361.0	8134.0	45475.0	
1	775635.0	8409.0	45453.0	
2	775132.0	8634.0	45373.0	
3	772351.0	8970.0	45293.0	
4	770821.0	9359.0	45214.0	
5	768649.0	9465.0	45084.0	
6	766778.0	9595.0	44956.0	
7	765754.0	9802.0	44907.0	
8	764875.0	10114.0	44875.0	
9	763447.0	10466.0	44791.0	

	onVentilatorCurrently	...	totalTestResults	lastModified	\
0	2802.0	...	363825123	2021-03-07T24:00:00Z	
1	2811.0	...	362655064	2021-03-06T24:00:00Z	
2	2889.0	...	361224072	2021-03-05T24:00:00Z	
3	2973.0	...	359479655	2021-03-04T24:00:00Z	
4	3094.0	...	357888671	2021-03-03T24:00:00Z	
5	3169.0	...	356481876	2021-03-02T24:00:00Z	
6	3171.0	...	355138357	2021-03-01T24:00:00Z	
7	3245.0	...	353983917	2021-02-28T24:00:00Z	
8	3335.0	...	352575495	2021-02-27T24:00:00Z	
9	3466.0	...	350920316	2021-02-26T24:00:00Z	

	recovered	total	posNeg	deathIncrease	hospitalizedIncrease	\
0	None	0	0	842	726	
1	None	0	0	1680	503	
2	None	0	0	2221	2781	
3	None	0	0	1743	1530	
4	None	0	0	2449	2172	
5	None	0	0	1728	1871	
6	None	0	0	1241	1024	
7	None	0	0	1051	879	
8	None	0	0	1847	1428	
9	None	0	0	2141	1868	

	negativeIncrease	positiveIncrease	totalTestResultsIncrease
0	131835	41835	1170059
1	143835	60015	1430992
2	271917	68787	1744417
3	177957	65487	1590984
4	267001	66836	1406795
5	255779	54248	1343519
6	118077	48092	1154440
7	203599	54349	1408422
8	205090	71245	1655179
9	276829	74857	1803309

[10 rows x 24 columns]

```
[573]: df_clean = df.dropna(subset=['positive', 'negative', 'pending', 'hospitalizedCurrently', 'hospitalizedCumulative', 'inIcuCurrently', 'inIcuCumulative'], how='any')
print(df_clean)
```

	date	states	positive	negative	pending	hospitalizedCurrently	\
0	20210307	56	28756489.0	74582825.0	11808.0	40199.0	
1	20210306	56	28714654.0	74450990.0	11783.0	41401.0	
2	20210305	56	28654639.0	74307155.0	12213.0	42541.0	
3	20210304	56	28585852.0	74035238.0	12405.0	44172.0	
4	20210303	56	28520365.0	73857281.0	11778.0	45462.0	
..	...	...	...	...	...	...	
342	20200330	56	171965.0	416268.0	65369.0	15892.0	
343	20200329	56	150826.0	366103.0	65404.0	14055.0	
344	20200328	56	131143.0	324271.0	65698.0	12393.0	
345	20200327	56	111358.0	255808.0	59795.0	10887.0	
346	20200326	56	92143.0	210652.0	60131.0	7805.0	

	hospitalizedCumulative	inIcuCurrently	inIcuCumulative	\
0	776361.0	8134.0	45475.0	
1	775635.0	8409.0	45453.0	
2	775132.0	8634.0	45373.0	
3	772351.0	8970.0	45293.0	
4	770821.0	9359.0	45214.0	
..	...	...	...	
342	5208.0	3087.0	187.0	
343	4624.0	2456.0	156.0	
344	4065.0	2174.0	140.0	
345	3383.0	1792.0	124.0	
346	2606.0	1299.0	91.0	

	onVentilatorCurrently	...	totalTestResults	lastModified	\
0	2802.0	...	363825123	2021-03-07T24:00:00Z	
1	2811.0	...	362655064	2021-03-06T24:00:00Z	

2	2889.0	...	361224072	2021-03-05T24:00:00Z
3	2973.0	...	359479655	2021-03-04T24:00:00Z
4	3094.0	...	357888671	2021-03-03T24:00:00Z
..	...	...	...	...
342	449.0	...	1068981	2020-03-30T24:00:00Z
343	433.0	...	968389	2020-03-29T24:00:00Z
344	390.0	...	881415	2020-03-28T24:00:00Z
345	293.0	...	769985	2020-03-27T24:00:00Z
346	258.0	...	668725	2020-03-26T24:00:00Z

	recovered	total	posNeg	deathIncrease	hospitalizedIncrease	\
0	None	0	0	842	726	
1	None	0	0	1680	503	
2	None	0	0	2221	2781	
3	None	0	0	1743	1530	
4	None	0	0	2449	2172	
..	...	...	...	...	...	
342	None	0	0	584	584	
343	None	0	0	506	559	
344	None	0	0	550	682	
345	None	0	0	411	777	
346	None	0	0	313	655	

	negativeIncrease	positiveIncrease	totalTestResultsIncrease
0	131835	41835	1170059
1	143835	60015	1430992
2	271917	68787	1744417
3	177957	65487	1590984
4	267001	66836	1406795
..	...	...	...
342	50165	21139	100592
343	41832	19683	86974
344	68463	19785	111430
345	45156	19215	101260
346	42359	17592	106945

[347 rows x 24 columns]

```
[576]: # Write the covid web DataFrame to a SQLite table
corona_df_clean.to_sql("cleaneddfweb", conn, if_exists="replace")
```

[576]: 232

```
[578]: # pushing Covid API Dataframe into Sql Table
df_clean.to_sql("cleanedAPI", conn, if_exists="replace")
```

[578]: 347

1.1 Join covidcountries and Cleaneddfweb tables to get death values by countries

```
[580]: sql = '''
        SELECT cc.Country,cc.Covidcases,cf.Deaths
        FROM covidCountries AS cc
             INNER JOIN cleaneddfweb AS cf ON cf.Country = cc.Country
        ORDER BY cc.Country

        '''

cursor.execute(sql)

# Fetching rows from the result table
result = cursor.fetchall()
for row in result:
    print(row)
```

```
('Afghanistan', 6828.0, '7998')
('Albania', 7158.3216, '3605')
('Algeria', 24044.6935, '6881')
('Andorra', 9758.0281, '159')
('Angola', 249.6712, '1937')
('Antigua and Barbuda', 233.2644, '146')
('Argentina', 27772.9672, '130663')
('Armenia', 15190.1073, '8777')
('Australia', 112560.8928, '25236')
('Austria', 248548.0663, '22534')
('Azerbaijan', 11723.720000000001, '10353')
('Bahamas', 495.63800000000003, '849')
('Bahrain', 19019.5775, '1536')
('Bangladesh', 4154.0413, '29499')
('Barbados', 977.6507, '593')
('Belarus', 17185.6632, '7118')
('Belgium', 379608.8333, '34339')
('Belize', 76.6507, '688')
('Benin', 403.62350000000004, '163')
('Bhutan', 227.9478, '21')
('Bolivia', 3292.1211000000003, '22387')
('Bosnia and Herzegovina', 12729.595000000001, '16392')
('Botswana', 114.35640000000001, '2801')
('Brazil', 228935.8397, '702116')
('Brunei', 3559.263, '179')
('Bulgaria', 11035.2197, '38700')
('Burkina Faso', 6972.6767, '400')
('Burundi', 72.5458, '15')
('Cabo Verde', 135.497, '417')
('Cambodia', 2884.4667, '3056')
```

('Cameroon', 9391.3501, '1974')
 ('Canada', 283292.1318, '55282')
 ('Central African Republic', 165.5505, '113')
 ('Chad', 213.1864, '194')
 ('Chile', 85974.7819, '62730')
 ('Colombia', 30892.2736, '142727')
 ('Costa Rica', 9321.9955, '9372')
 ('Croatia', 23376.3, '18752')
 ('Cuba', 6581.0, '8530')
 ('Cyprus', 8656.5563, '1451')
 ('Denmark', 99170.84909999999, '9693')
 ('Djibouti', 1709.4153999999999, '189')
 ('Dominica', 221.04399999999998, '74')
 ('Dominican Republic', 33230.573000000004, '4384')
 ('Ecuador', 79100.9854, '36050')
 ('Egypt', 26103.0, '24830')
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 ('Grenada', 161.4375, '238')
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 ('Saint Lucia', 201.9305, '410')
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 ('Seychelles', 299.8124, '172')
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 ('Sri Lanka', 3948.0, '16907')
 ('Sudan', 282.0804, '5046')
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 ('Switzerland', 445000.0457, '14170')
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 ('Tanzania', 536.5198, '846')
 ('Thailand', 43873.0, '34715')
 ('Timor-Leste', 145.853322, '138')
 ('Togo', 1058.4443, '290')
 ('Trinidad and Tobago', 2070.4692999999997, '4390')
 ('Tunisia', 10298.0, '29423')
 ('Turkey', 526623.2069999999, '101419')
 ('Uganda', 905.0, '3632')
 ('Ukraine', 25373.545, '109920')
 ('United Arab Emirates', 38103.0, '2349')
 ('United Kingdom', 923997.1052, '232112')
 ('Uruguay', 8135.7114, '7682')
 ('Uzbekistan', 7746.9628, '1016')
 ('Venezuela', 3375.8341, '5856')
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 ('Yemen', 68.069115, '2159')
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 ('Zimbabwe', 208.0, '5740')

1.2 Get Year = 2021 data for covid i.e. positive, negative and pending reports for covid patient.

```
[585]: sql = '''
        SELECT substr(date,1,4) as_
        ↪APIDate,positive,negative,pending,hospitalizedCurrently
        FROM cleanedAPI AS cf
        WHERE substr(date,1,4) = '2021'

        '''

cursor.execute(sql)

# Fetching rows from the result table
result = cursor.fetchall()
for row in result:
    print(row)
```

```
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('2021', 27821578.0, 71507723.0, 9408.0, 58222.0)
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('2021', 27357332.0, 70272654.0, 11529.0, 69207.0)
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('2021', 27165660.0, 69782378.0, 11981.0, 74208.0)
('2021', 27063243.0, 69522254.0, 12079.0, 76979.0)
('2021', 26968049.0, 69366393.0, 10516.0, 79179.0)
('2021', 26875063.0, 69029225.0, 12114.0, 80055.0)
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```

```
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( '2021', 24629099.0, 64823753.0, 11247.0, 116264.0)
( '2021', 24438184.0, 64539361.0, 10907.0, 119949.0)
( '2021', 24251909.0, 64271666.0, 10529.0, 122754.0)
( '2021', 24062706.0, 63929216.0, 11503.0, 123910.0)
( '2021', 23916080.0, 63599369.0, 11477.0, 123848.0)
( '2021', 23765288.0, 63406544.0, 11404.0, 124387.0)
( '2021', 23578070.0, 63128702.0, 11334.0, 126139.0)
( '2021', 23359985.0, 62902360.0, 11948.0, 127235.0)
( '2021', 23113531.0, 62491363.0, 12138.0, 128947.0)
( '2021', 22887915.0, 62185469.0, 10942.0, 130391.0)
( '2021', 22663424.0, 61947584.0, 9181.0, 131326.0)
( '2021', 22445404.0, 61503591.0, 11039.0, 129793.0)
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( '2021', 22021417.0, 61022599.0, 10850.0, 130781.0)
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( '2021', 21456928.0, 60412132.0, 11612.0, 132370.0)
( '2021', 21184885.0, 60165554.0, 10062.0, 132474.0)
( '2021', 20934701.0, 59902466.0, 8008.0, 131195.0)
( '2021', 20715626.0, 59422210.0, 9463.0, 128210.0)
( '2021', 20536055.0, 59273596.0, 9407.0, 125562.0)
( '2021', 20327598.0, 59058013.0, 9356.0, 123614.0)
( '2021', 20047280.0, 58836221.0, 11506.0, 125047.0)
```

1.3 Grouping covid metrics by year.

```
[589]: sql = '''
        SELECT substr(date,1,4) as APIDate,SUM(positive) as positivecases,
        SUM(negative) as negativecases,AVG(pending) as_
        pendingcases,AVG(hospitalizedCurrently) as hospitalized
        FROM cleanedAPI AS cf
        GROUP BY substr(date,1,4)

        '''
```



```
cursor.execute(sql)

# Fetching rows from the result table
result = cursor.fetchall()
for row in result:
    print(row)
```

```
('2020', 1711472650.0, 6828680724.0, 9640.38078291815, 52113.55516014235)
('2021', 1689066577.0, 4443463114.0, 11093.151515151516, 90629.54545454546)
```

```
[ ]:
```